

Packet Tracer

Modify Single-Area OSPFv2

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
R1	S0/0/0	172.16.3.1	255.255.255.252	N/A
R1	S0/0/1	192.168.10.5	255.255.255.252	N/A
R2	G0/0	172.16.2.1	255.255.255.0	N/A
R2	S0/0/0	172.16.3.2	255.255.255.252	N/A
R2	S0/0/1	192.168.10.9	255.255.255.252	N/A
R2	S0/1/0	209.165.200.225	255.255.255.224	N/A
R3	G0/0	192.168.1.1	255.255.255.0	N/A
R3	S0/0/0	192.168.10.6	255.255.255.252	N/A
R3	S0/0/1	192.168.10.10	255.255.255.252	N/A
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

- Part 1: Modify OSPF Default Settings
- Part 2: Verify Connectivity

Scenario

In this activity, OSPF is already configured and all end devices currently have full connectivity. You will modify the default OSPF routing configurations by changing the hello and dead timers and adjusting the bandwidth of a link. Then you will verify that full connectivity is restored for all end devices.

Instructions

Part 1: Modify OSPF Default Settings

Step 1: Test connectivity between all end devices.

Before modifying the OSPF settings, verify that all PCs can ping the web server and each other.

All PC can ping to web server

Step 2: Adjust the hello and dead timers between R1 and R2.

a. Enter the following commands on R1.

Open configuration window

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf hello-interval 15
R1(config-if)# ip ospf dead-interval 60
```

Ip ospf hello-interval 15 เป็นการกำหนดให้ส่ง hello packet ให้กับเพื่อนบ้านทุก 15 วินาที

Ip ospf dead-interval 60 กำหนดว่าหาก เพื่อนบ้านไม่ตอบกลับ hello packet ภายใน 60 วินาที จะนับว่าเพื่อนบ้านตัวนั้น down

b. After a short period of time, the OSPF connection with R2 will fail, as shown in the router output.

```
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Dead timer expired
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Interface down or detached
```

จากเพื่อนบ้าน (209.165.200.225) เปลี่ยนสถานะจาก full to down เพราะหมดเวลารับ hello packet (dead timer expired) จากที่กำหนดไว้

Both sides of the connection need to have the same timer values in order for the adjacency to be maintained. Identify the interface on R2 that is connected to R1. Adjust the timers on the R2 interface to match the settings on R1.

After a brief period of time you should see a status message that indicates that the OSPF adjacency has been reestablished.

```
00:21:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL,
Loading Done
```

Step 3: Adjust the bandwidth setting on R1.

a. Trace the path between PC1 and the web server located at 64.100.1.2. Notice that the path from PC1 to 64.100.1.2 is routed through R2. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:

  1      1 ms      0 ms      8 ms    172.16.1.1
  2      0 ms      1 ms      0 ms    172.16.3.2
```

```

3      1 ms      9 ms      2 ms  209.165.200.226
4      *         1 ms      0 ms  64.100.1.2

```

Trace complete.

b. On the R1 Serial 0/0/0 interface, set the bandwidth to 64 Kb/s. This does not change the actual port speed, only the metric that the OSPF process on R1 will use to calculate best routes.

```
R1(config-if)# bandwidth 64
```

กำหนดค่า cost โดย bandwidth เพื่อให้ ospf รู้ค่าน้ำหนักใช้ในการคำนวณหา SPF ยิ่ง

c. Trace the path between PC1 and the web server located at 64.100.1.2. Notice that the path from PC1 to 64.100.1.2 is redirected through R3. OSPF prefers the lower cost path.

```

C:\> tracert 64.100.1.2
Tracing route to 64.100.1.2 over a maximum of 30 hops:

```

```

1      1 ms      0 ms      3 ms  172.16.1.1
2      8 ms      1 ms      1 ms  192.168.10.6
3      2 ms      0 ms      2 ms  172.16.3.2
4      2 ms      3 ms      1 ms  209.165.200.226
5      2 ms     11 ms     11 ms  64.100.1.2

```

Trace complete.

Close configuration window

Part 2: Verify Connectivity

Verify that all PCs can ping the web server and each other.

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